

Conventional, Organic, Biodynamic & Natural Wine

What the labels actually mean — and what they don't · kazmicellars.com

These four terms represent a spectrum of winemaking philosophy — from heavily interventionist to radically hands-off. They're often confused, misrepresented, and used as marketing. Here's what each actually means, what the certifications require, and what ends up in your glass.

CONVENTIONAL WINEMAKING

What it means	Most wine in the world is made conventionally. There are no restrictions on what can be used in the vineyard or the winery, within national food safety regulations.
In the vineyard	Synthetic pesticides, herbicides, fungicides, and fertilizers are all permitted. Irrigation is common in New World regions. Mechanized harvesting is standard.
In the winery	Commercial yeasts are standard. A wide range of additives is permitted including sulfites (SO ₂), tannin powder, acidification, deacidification, water addition (in some regions), mega purple (color concentrate), and dozens of fining agents. Oak chips and staves can be used instead of barrels.
Sulfites (SO₂)	Added as a preservative and antioxidant at various stages. Conventional wines may contain up to 350mg/L (sweet whites). Dry red wines average 80–150mg/L. All wines with more than 10mg/L must state 'Contains Sulfites' on the label in most markets.
Quality	Conventional winemaking produces wine across the entire quality spectrum — from the cheapest supermarket bottle to some of the world's greatest estates. Conventional ≠ inferior.

ORGANIC WINEMAKING

What it means	Organic certification focuses primarily on the vineyard, not the winery. It prohibits synthetic pesticides, herbicides, and fertilizers. It does NOT ban sulfites (despite common belief) in most jurisdictions.
Key certifications	USDA Organic (USA): 'Organic Wine' must have no added sulfites and less than 10mg/L total. 'Made with Organic Grapes' allows up to 100mg/L sulfites. Ecocert / AB (France/Europe): organic farming but sulfites are permitted (up to 100mg/L red, 150mg/L white). Demeter (see Biodynamic) covers both organic and biodynamic standards.
In the vineyard	Copper sulfate and sulfur sprays are the main tools for disease control — both are naturally derived. Cover crops, composts, and natural fertilizers are used.
Common misconception	'Organic' does not mean no sulfites in Europe or most of the world. Only US 'Organic Wine' (not 'Made with Organic Grapes') prohibits added sulfites.
Why it matters	Less chemical residue in the soil, better soil microbiome diversity, reduced environmental impact. Many great estates farm organically without certifying — the certification costs money and requires years of transition.

BIODYNAMIC WINEMAKING

What it means	Biodynamics goes well beyond organic farming. It treats the vineyard as a self-sustaining ecosystem — incorporating lunar cycles, herbal preparations, and a holistic philosophy developed by Rudolf Steiner in 1924. Certified by Demeter.
The lunar calendar	Biodynamic farmers follow a calendar that categorizes each day as a Root day, Flower day, Fruit day, or Leaf day. Pruning, harvesting, and bottling are scheduled according to this calendar. Tastings on Root days are said to show less expressiveness; Fruit days are considered optimal.
The preparations	Nine herbal preparations (numbered 500–508) are central to biodynamics. Preparation 500 (horn manure): cow dung fermented in a cow horn, buried over winter, then stirred in water and sprayed on soil at incredibly dilute concentrations. Preparation 501 (horn silica): quartz powder buried in summer, then sprayed on vines at sunrise. These preparations are used in homeopathic quantities — skeptics call it pseudoscience; practitioners point to vine health and wine quality.
Demeter certification	The most rigorous wine certification in the world. Requires the entire farm to be biodynamic (not just the vineyard), mandatory use of the preparations, and compliance with Demeter's winery standards on sulfite levels.
Who uses it	Many of the world's most celebrated estates are biodynamic: Domaine de la Romanée-Conti, Domaine Leroy, Domaine Leflaive, Zind-Humbrecht, Nicolas Joly (Coulée de Serrant), Tablas Creek (Paso Robles). Their commitment suggests something is working.

NATURAL WINE

What it means	The most contested term in wine. Natural wine has no universal legal definition. Generally understood to mean: organic or biodynamic farming + wild yeast fermentation + no or minimal sulfite addition + unfinned + unfiltered. Beyond that, definitions vary widely.
What's forbidden (generally)	Added commercial yeast, acidification, deacidification, fining agents, filtration, and significant SO ₂ additions. Also typically: mega purple, water addition, oak chips, and any intervention beyond basic cellar hygiene.
The sulfite question	This is the core debate. Some natural wine producers add no sulfites at all; others add a minimal amount at bottling (typically <30mg/L vs up to 350mg/L conventional). Zero-sulfite wines are microbiologically risky and unstable — they must be stored and transported cold and drunk promptly.
Wild yeast fermentation	Instead of inoculating with commercial yeast, the wine ferments with native yeasts from the vineyard and winery. This is slower, less predictable, and creates more complexity — but also more risk of stuck fermentation or off-flavors.
Cloudiness & funkiness	Many natural wines are cloudy (unfinned/unfiltered) and show some degree of 'funk' from native yeasts and minimal SO ₂ . This is intentional. However, it can sometimes mask genuine faults. Not all cloudiness is intentional; not all funk is desirable.
Quality spectrum	Natural wine ranges from transcendent (some of the most exciting, individual wines being made today) to genuinely faulty (unstable, oxidized, or simply unpleasant). The label alone doesn't guarantee quality — as with all wine, you need to know the producer.

■ Many great estates farm organically or biodynamically without certifying — because the certification takes 3+ years and costs money. Absence of certification ≠ conventional farming. Ask the producer, or look for estates known to farm this way: DRC, Leroy, Leflaive in Burgundy; Pontet-Canet and Palmer in Bordeaux; Movia in Slovenia; Gravner and Radikon in Friuli.